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MID TERM EXAMINATION II
 B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
 Second Semester, April 2026

Paper Code: ES-102**Subject: Programming in C****Time: 1½ Hrs.****Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1. (a)	What is Generic Pointer? Explain with suitable example.	2	C03
(b)	Explain self-referential structure.	2	C03
(c)	Explain command line arguments in C with the help of suitable example.	2	C02
(d)	Sort the following numbers stepwise using Insertion sort: 9 1 4 15 6 13 2 8	2	C04
(e)	Write any two library functions of the following header files: 1. stdio.h 2. stdlib.h	2	C02
2. (a)	What is the difference between Structure and Union? Explain with suitable example.	5	C03
(b)	Write a C program to implement Bubble sort.	5	C02
3. (a)	What is Dynamic Memory Allocation? Explain the difference between malloc and calloc .	5	C03
(b)	What is Binary Search? Search an item 12 from the following sorted data elements using Binary Search: 7 9 12 15 20 23 28	5	C04
4. (a)	Write a C program to create a file using write mode and display the contents using read mode.	5	C03
(b)	Explain Selection Sort. Sort the following numbers using Selection Sort: 87 43 54 22 98 76 65	5	C04

**** End of Question Paper ****